CIVITAS
Sustainable and smart mobility for all
2030

CIVITAS indicators

Energy efficiency of freight transport (ENG_EF_EE2)

DOMAIN

Transport	Environment	**Energy**	Society	Economy

TOPIC	**Energy efficiency**	
IMPACT	**Energy efficiency of freight transport** *Increasing the energy efficiency of freight transport*	**ENG_EF**

Category

Key indicator	Supplementary indicator	State indicator

CONTEXT AND RELEVANCE

Transport activity is a major source of pollution and greenhouse gas emissions, primarily due to energy consumption. Improving energy efficiency through technological advancements allows vehicles to use less fuel per kilometre, reducing CO_2 emissions and other pollutants. Efficiency can also be increased through higher load factors, smoother driving and better traffic management.

This indicator measures the energy efficiency of freight transport, expressed as energy consumption per tonne-kilometre moved. It captures the amount of energy required to generate a given level of freight transport activity. The indicator is relevant when policy actions aim to improve vehicle efficiency, shift travel to more energy-efficient modes, improve load factors, or reduce congestion and stop-and-go traffic. **A successful action is reflected in a <u>LOWER</u> value of the indicator, indicating that less energy is required per tonne-kilometre travelled.**

DESCRIPTION

This indicator measures the energy efficiency of freight transport by calculating the ratio of energy consumed to the amount of freight transported, expressed in tonne-kilometres. The unit of measurement of the indicator is **tonnes of oil equivalent per tonne-kilometre (toe/tkm)**.

METHOD OF CALCULATION AND INPUTS

This indicator draws from an urban transport model to obtain the energy demand by fuel type and the total freight volume in the pilot area. I**f a city transport model exists and provides an estimation of energy demanded by all relevant modes of transport for all relevant fuel types**, this method has limited complexity. Furthermore, assuming that the model is reliable, results have a good level of significance. However, **if no urban transport model exists**, complexity increases significantly, as developing such a model is time-consuming and technically demanding.

The supporting tool computes the indicator endogenously, using the input data along with predefined parameters coded within the tool.

Method 1	
Energy demand and freight volumes drawn from an urban transport model	Significance: **0.50**

INPUTS

The following information is needed to compute the indicator:

a) **The modelled energy demand of freight vehicles in the pilot area by fuel type**, over the period covered by the urban transport model. The model must include all fuel types relevant to the pilot area; otherwise, the coverage of the indicator will be incomplete.

b) **The modelled volume of freight transported in the pilot area** over the same period.

Urban transport models may cover various timeframes, e.g., one peak hour, two peak hours, one peak and one off-peak hour, or an average day. Regardless of the period, the modelled energy demand and freight volume are used directly as inputs for this indicator. Scaling to annual values is not required.

METHOD OF CALCULATION

The indicator is calculated **endogenously** by the supporting tool. The user must provide the energy demand of freight vehicles and the freight volume transported in the pilot area, extracted from the transport model. The tool then computes the indicator using the equation and parameters described in the section below.

EQUATIONS

The value of the indicator is computed as:

$$\boldsymbol{FrEngEff} = \frac{\sum_e(\boldsymbol{ModEngDem^e} * \boldsymbol{ConvFact^e})}{\boldsymbol{ModFrActv}}$$

Where:

$ModEngDem^e$ = Energy demand by freight vehicles for fuel type e in the modelled period, in million litres (gasoline, diesel, biodiesel, bioethanol), million kilograms (LPG, CNG, biomethane, hydrogen) and million kWh (electricity).

$ConvFact^e$ = Conversion factor from the native unit to toe for fuel type e.

$ModFrActv$ = Total freight transport activity in the modelled period, in tonne-kilometres.

The conversion factors $\boldsymbol{ConvFact^e}$ from native units to toe are predefined values in the supporting tool. The factors are summarized in the following table.

	Conversion factor	*Unit*
Gasoline	776	$toe/(Mio * l)$
Diesel	857	$toe/(Mio * l)$
Biodiesel	780	$toe/(Mio * l)$
Bioethanol	503	$toe/(Mio * l)$
LPG	561	$toe/(Mio * kg)$
Biomethane	801	$toe/(Mio * kg)$
CNG	1195	$toe/(Mio * kg)$
Battery electric	86	$toe/(Mio * kWh)$
Fuel cells	2866	$toe/(Mio * kg)$

ALTERNATIVE INDICATORS

While this indicator measures the energy efficiency of freight transport in the pilot area, indicator **ENG_EF_EE1** measures the **energy efficiency of passenger transport**.